

Reference angles and trigonometric functions



- 1 a Quadrant 2 b 45° c Positive d $\sin 45^\circ$
- 2 a Quadrant 3 b 60° c Negative d $-\cos 60^\circ$
- 3 a Quadrant 3 b 30° c Positive d $\tan 30^\circ$
- 4 a Quadrant 3 b $\frac{\pi}{6}$ c Negative d $-\sin \frac{\pi}{6}$
- 5 a Quadrant 4 b $\frac{\pi}{6}$ c Positive d $\cos \frac{\pi}{6}$
- 6 a Quadrant 3 b $\frac{\pi}{4}$ c Positive d $\tan \frac{\pi}{4}$
- 7 a Quadrant 4 b $\frac{\pi}{6}$ c Negative d $-\tan \frac{\pi}{6}$
- 8 a Quadrant 4 b $\frac{\pi}{4}$ c Positive d $\cos \frac{\pi}{4}$
- 9 a Quadrant 3 b $\frac{\pi}{3}$ c Negative d $-\sin \frac{\pi}{3}$
- 10 a Quadrant 3 b $\frac{\pi}{3}$ c Negative d $-\cos \frac{\pi}{3}$
- 11 a Quadrant 4 b $\frac{\pi}{4}$ c Negative d $-\tan \frac{\pi}{4}$

Exact values of trigonometric function



12

a $(0, 1)$

c 1

b 0

d

i No ii No iii Yes iv Yes

13

a $(1, 0)$

c 0

b 1

d

i Yes ii Yes iii No iv No

14

a $(0, -1)$

c -1

b 0

d

i Yes ii No iii No iv Yes

15

a $(-1, 0)$

c 0

b -1

d

i No ii Yes iii Yes iv No

16

a $(1, 0)$

c 0

b 1

d

i No ii Yes iii No iv Yes

17

a $(-1, 0)$

c 0

b -1

d

i Yes ii Yes iii No iv No

18

a $-\frac{1}{2}$

e $\frac{2\sqrt{3}}{3}$

i $-\frac{1}{\sqrt{2}}$

b $\frac{\sqrt{3}}{2}$

f $-\sqrt{3}$

j 2

c $-\frac{\sqrt{3}}{3}$

g $\frac{1}{2}$

k $-\frac{2\sqrt{3}}{3}$

d -2

h $-\frac{\sqrt{3}}{2}$

l $-\sqrt{3}$



19

a $\frac{\sqrt{3}}{2}$

e -2

b $-\frac{1}{2}$

f $-\frac{\sqrt{3}}{3}$

c $-\sqrt{3}$

d $\frac{2\sqrt{3}}{3}$

20

a i 60° ii $-\frac{1}{2}$

b i 0 ii 0

c i $\frac{\pi}{3}$ ii $-\frac{1}{2}$

21

a 1

e Undefined

i $\frac{2\sqrt{3}}{3}$

m 0

q Undefined

b 1

f Undefined

j $\sqrt{2}$

n -1

r Undefined

c -1

g $\sqrt{3}$

k $\sqrt{2}$

o Undefined

d -1

h $\frac{2\sqrt{3}}{3}$

l 1

p 0

22

a 1

e $\frac{1}{2}$

i $\frac{1}{\sqrt{3}}$

m $-\frac{1}{\sqrt{3}}$

b $\frac{-1}{2}$

f $-\frac{1}{\sqrt{2}}$

j $-\frac{\sqrt{3}}{2}$

n Undefined

c $\frac{1}{2}$

g $-\frac{1}{2}$

k $\frac{1}{2}$

o Undefined

d 1

h $-\frac{\sqrt{3}}{2}$

l -1

p Undefined

23

a $-\frac{1}{4}$

b $-\frac{\sqrt{3}}{4}$

c -2

d $-2\sqrt{3}$